

ALTERNATIVES TO ENUCLEATION IN THE MANAGEMENT OF CHOROIDAL MELANOMA

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Abstract

Over the past 15 years 140 patients with choroidal melanoma have been treated by local surgical resection of the tumour. The method appears worth considering for tumours up to 15 mm in diameter not involving more than one-third of the ciliary body. Sixty percent of patients retained useful vision and 25% good vision. Eighty-two percent retained a cosmetically satisfactory eye. The total tumour-related mortality in this group of patients was 9%, almost all of the 12 patients dying having had tumours larger than would now be recommended for local resection. The five-year survival rate among patients undergoing local resection was 84%.

Low-energy long-exposure laser has been used to treat a small number of choroidal tumours. The technique appears useful for tumours up to 10 mm in diameter and not more than 3 mm in thickness. Ruthenium plaque therapy has also been used in some cases and appears a suitable method of treatment for tumours up to 10 mm in diameter and not more than 6 mm in thickness.

The options for conservative management of choroidal tumours are greater with small tumours than with large ones except in only eyes, tumours of over 15 mm in diameter are best treated with enucleation.

The long-term survival after all forms of conservative management has not yet been established. At present the use of controversial techniques such as local resection should be restricted to a few centres so that adequate statistical evaluation of the method may be made.

Key words: Choroidal melanoma, conservative management, local resection, laser treatment, radioactive plaque.

Over the past 15 years we in Glasgow have had an interest in the conservative management of ocular tumours. Currently up to three new patients with ocular tumours are referred to us each week from various parts of the UK for investigation and management. The majority of these patients are suffering from choroidal melanomas, although retinoblastomas, secondary tumours and tumours of the external eye and adnexae are also seen.

In relation to the choroidal melanomas, our main interest has been in the assessment of local surgical resection in their management, and to date approximately 140 patients with choroidal

melanomas have been managed in this way. Another group of patients (numbering 22) has been treated by radiotherapy using radioactive cobalt or ruthenium plaques, while still other patients have been managed by observation alone or by treatment with low-energy long-exposure laser, occasionally coupled with haematoporphyrin-derivative photosensitization.

As all ophthalmologists know, there is at present an unresolved controversy about the best management of patients with choroidal melanoma in relation to subsequent mortality or survival. Overall the death rate after enucleation for choroidal melanoma averages some 50% in

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five years in most series,^{1,2} and is related to such factors as cell type,^{3,4} tumour volume,^{5,6} the presence of extrascleral spread and possibly manipulation of the eye during enucleation. This last factor has certainly been invoked to explain the peak mortality seen one to two years after enucleation in the material reported by Zimmermann (and seen also in other published series).⁷

A recent multifactorial analysis of factors influencing prognosis in choroidal melanoma has shown clearly that the most important factor is the size of the tumour at enucleation.⁸ Other factors such as the presence of epithelioid cells or extraocular extension relate directly to tumour volume and on their own are less significant than measurements of tumour volume.

There is no doubt that the death rate after enucleation of an eye for a small tumour is low,^{5,6} but so too is the mortality in similar patients managed by observation alone.⁹ The unresolved controversy has led to diametrically opposed views on the management of these tumours, some authors recommending early enucleation of eyes with small tumours¹⁰ while the opposite view has also been expressed, namely, that very small tumours are sometimes naevi or relatively benign lesions and that observation alone is the best method of management.¹¹ Against observation as a method of management is the suggestion that resorting to enucleation only if a tumour enlarges may convert a small tumour with a good prognosis to a larger one with a poorer prognosis.¹²

Certainly tumours vary greatly in their behaviour and we ourselves have seen some tumours which have scarcely altered at all over many years and others which have increased rapidly in size over the course of a few weeks.

Some 15 or more years ago, influenced by studies that appeared to show no improvement in mortality after enucleation of eyes for uveal melanomas, one of us (WSF) began to explore the possibility of local surgical resection of choroidal melanomas as an alternative to enucleation. Isolated reports of such operations had already appeared in the literature,¹³⁻¹⁵ but there were obvious technical problems associated with this form of surgery, not the least of which were

the vascularity of the choroid and the vulnerability of the retina in terms of subsequent development of retinal detachment. The first of these problems can be circumvented by the use of systemic hypotensive anaesthesia and the second, in most instances, by a dissection which leaves the retina intact. The use of systemic hypotension to achieve haemostasis means, in general, that this form of surgery is contraindicated in elderly patients and in those with cardiovascular problems.

Avoidance of retinal detachment after local surgical resection of choroidal tumours is not always achieved, however, for a proportion of eyes undergoing local resection subsequently develop detachment, usually as a complication of postoperative vitreous haemorrhage and traction and not as a rule the result of a retinal tear. Indeed it has been our experience that retinal tears uncomplicated by vitreous haemorrhage do not as a rule lead to retinal detachment¹⁶ especially if the tear is situated over the surgical coloboma. Even where retinal detachment does occur, in many cases it can be dealt with using conventional surgical techniques including vitrectomy and silicone oil replacement when indicated. The incidence of retinal detachment in 140 cases of local resection of choroidal tumours was 19% (27 cases). Seven out of 17 operated on (41%) responded satisfactorily to retinal detachment surgery.

TECHNIQUES

The technique of local surgical resection has previously been reported,^{17,18} but basically the approach is via a half-thickness lamellar dissection of sclera over the tumour which has previously been localized by trans-pupillary fibreoptics illumination, which casts a sharp outline of the tumour on the overlying sclera. The choroid over the tumour is next exposed by removal of the peripheral portion of the deep scleral lamella. Healthy choroid some 3 to 4 mm from the tumour is incised into the subretinal space and the choroid containing the tumour is dissected free from the intact underlying retina by careful blunt dissection under the operating microscope. Occasionally the tumour is adherent

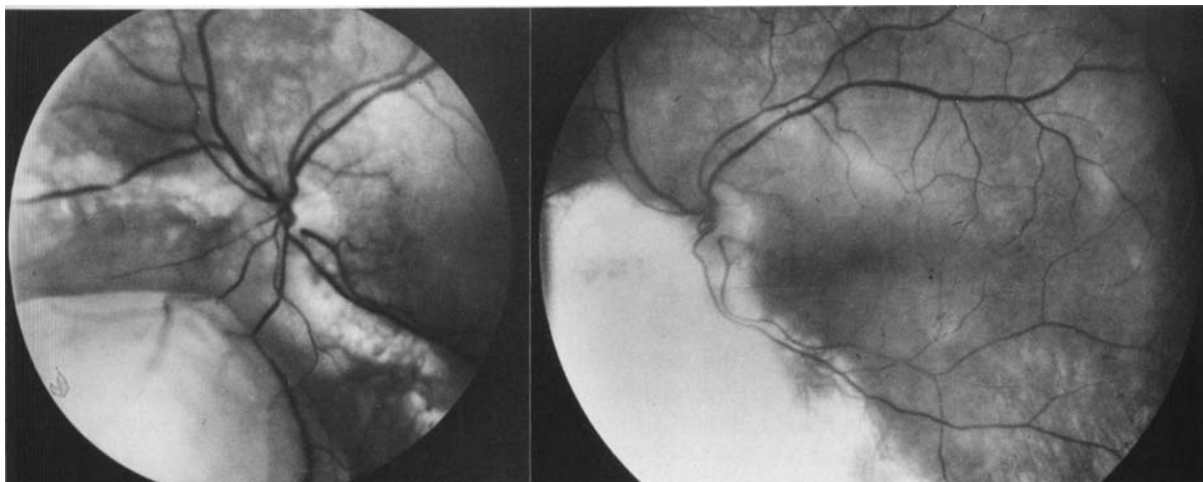


Figure 1: A posteriorly placed choroidal melanoma treated by local surgical resection. Proximity to the optic disc necessitated a modified approach to the tumour. (a) Preoperative appearance, (b) postoperative appearance. Visual acuity 6/9.

to the retina and, in these cases, the outer retina is removed with the tumour. Sometimes a full thickness retinal break occurs at this point, but as already indicated this does not as a rule lead to retinal detachment.

Modifications to the technique are required if the tumour is adjacent to the optic nerve head (Figure 1) if there is extrascleral extension or where the tumour involves a significant proportion of the ciliary body. In the latter instance removal of the lens may be necessary to prevent subluxation and visual loss from lens opacity

and, in these cases, an extensive anterior vitrectomy may also be required. Localized extrascleral extension demands a full thickness scleral resection of the involved sclera and healthy surround. Repair of the defect is currently achieved by the use of a half-thickness scleral graft from an uninvolved area of the same eye (Figure 2).

Preoperative circumvallation with laser is only used for smaller tumours unaccompanied by significant serous detachment of the retina. The size range of tumours thought suitable for local resection is from 10 to 15 mm in diameter.

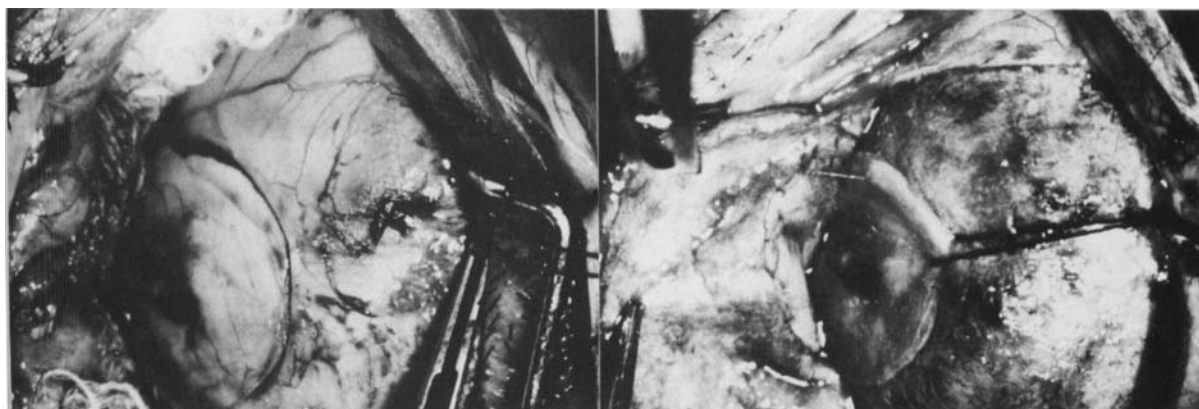


Figure 2: (a) Extrascleral extension from a choroidal tumour being locally resected from an only eye. (b) A full thickness scleral resection at the site of the extension is repaired with a half-thickness homo-transplant from the same eye.

Tumours of 16 mm or over in diameter are technically resectable, but this size of tumour carries a poor prognosis irrespective of the treatment adopted and, except in the case of only eyes, patients with this size of tumour are advised to undergo enucleation to avoid problems of secondary glaucoma and a painful blind eye.

RESULTS

What about the results of this form of surgery? First, as regards the affected eye, a good cosmetic result has been achieved in 110 of the first 134 choroidal resections (82%), useful vision (that is, useful field of vision) in 80 of 134 (60%), and good vision (6/6 to 6/18) in 33 of 134 (25%). In 43% of resections (58 of 134) vision was better than 6/60. In spite of earlier fears, the only case of overt orbital recurrence was in an early patient who had an attempted local resection in 1970 which was abandoned because of haemorrhage, neither the technique nor the anaesthesia having been refined at that stage. The tumour was large and epithelioid and, in spite of the orbital recurrence which was removed surgically, the patient survived 13 years before dying of liver metastases. Postoperatively a few eyes have had to be enucleated mainly because of traction retinal detachment not responding to treatment and often accompanied by uveitis and hypotony. As already indicated, however, the majority of patients retain a cosmetically acceptable eye and indeed the majority retain useful vision. Enucleation, however, is recommended for any eye in which visual recovery is uncertain and in which there is a possibility of residual tumour — 14% have fallen into this category. Enucleation for recurrent tumour has only been necessary in 12 cases (8.5%).

As regards mortality, 15 patients of the 140 have died (11%), but only 12 from probable metastases. Two patients have died of other tumours and one from other causes. In the absence of an autopsy, the actual cause of death in some of our patients must remain conjectural. It is relevant to note that, of the 12 patients dying from presumed or proven metastases, nine had tumours of over 16 mm in diameter and would not have met current criteria for local resection.

In addition, one of the remaining three patients had extrascleral spread at the time of surgery. The five-year postoperative survival rate for the 31 patients operated on prior to 1980 is 84% (five tumour-related deaths).

The number of operations carried out between 1970 and 1978 averaged about three per year, but since 1979 the number of local resections carried out has increased dramatically rising to more than 30 per year currently. The critical evaluation of this operation in terms of survival will be the tumour-related mortality in the large group of patients operated on during the last few years. At present we are comparing our mortality figures (matched for tumour size, cytological features and other factors) with eyes enucleated during the same period, and we have made preliminary arrangements to compare our results with those obtained in Boston using proton beam irradiation. We are cooperating with Professor Gamel in St Louis who is making an estimate of the likely survival of our cases had they been treated by enucleation based on certain cytological characteristics¹⁹ of the resected tumour. This estimated survival will be compared with the actual survival in our group of surgically resected patients.

At present we can say with reasonable confidence that the results of local resection in terms of mortality appear not to be worse and, indeed, may be better than enucleation and, of course, retention of the eye is a considerable bonus as far as the patient is concerned.

OTHER MANAGEMENT MODALITIES

What about other forms of management? Like others, we are prepared to observe and serially document small tumours which may be relatively benign. Where such tumours, usually in the size range up to 8 mm in diameter, show a documented increase in size or have done so prior to the patient being referred to us, we employ either local irradiation using ruthenium/rhodium plaque therapy,²⁰ or low-energy long-exposure laser,²¹ depending on the measured thickness of the tumour by ultrasonography. For thin tumours up to 3 mm in thickness laser treatment is used. Preliminary circumvallation with

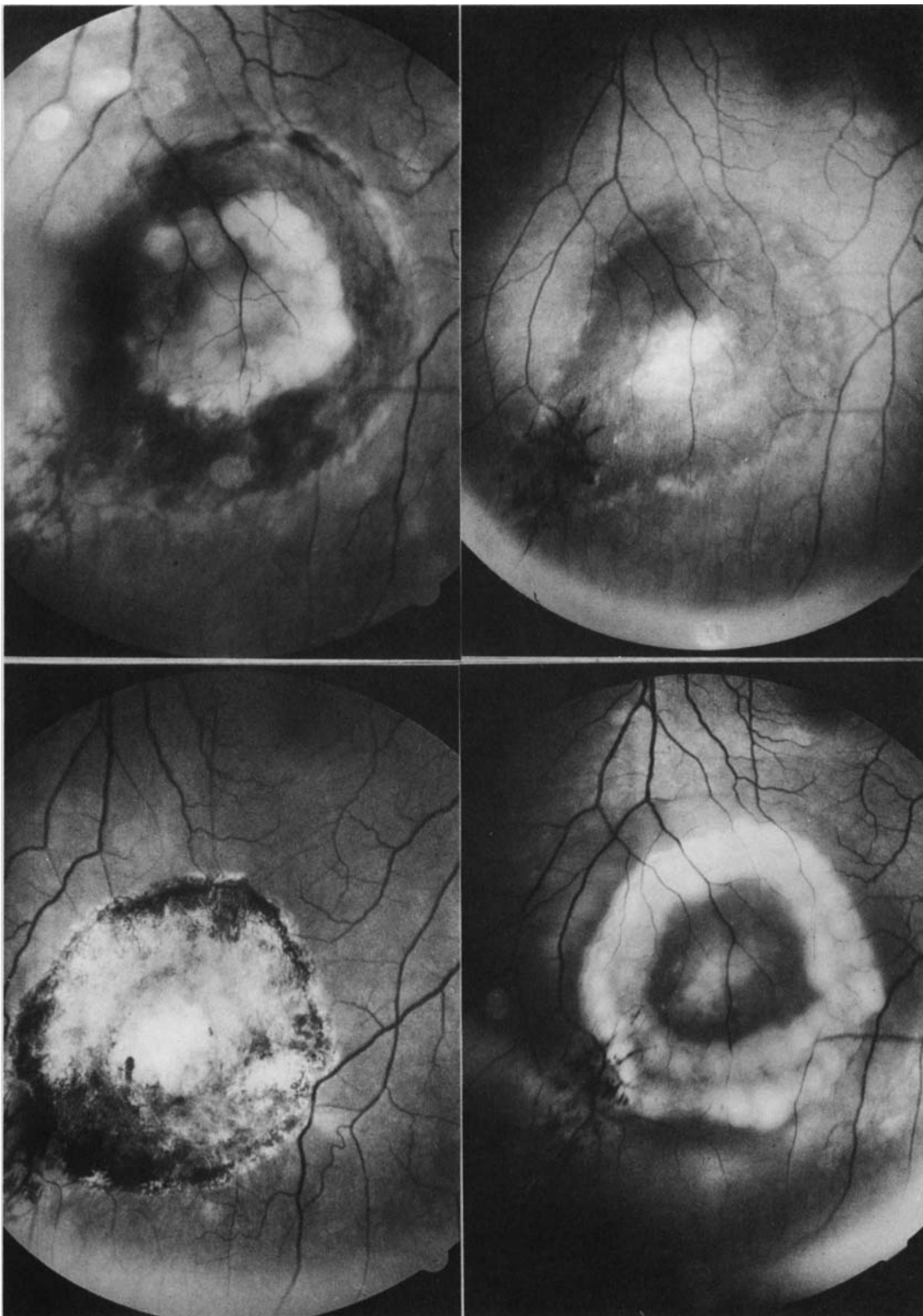


Figure 3: (a) Pre-treatment and (b) post-treatment appearances of a choroidal melanoma treated by a combination of argon green and krypton red laser.

argon green laser burns, characteristically 0.2 W for durations up to 20 seconds, are followed in subsequent treatment sessions with krypton red applications gradually working in from the periphery to the centre of the tumour. Such a regimen seems to be an effective treatment which avoids surgical intervention (Figure 3). The treatment has to be given in carefully fractionated doses if exudative retinal detachment or retinal traction or even neovascularization are to be avoided. So far we have treated 14 choroidal tumours in this way and, of these within the size range considered suitable (that is, up to 10 mm in diameter and up to 3 mm thick) 12 have responded well. In only one case has there been continued tumour growth, while in only one early case treated rather over-enthusiastically, vision was lost from traction detachment and neovascularization. Two patients with thicker

tumours too close to the optic disc for radioactive plaque therapy, and who refused enucleation, were treated with long-exposure laser after haematoporphyrin photosensitization. In both cases tumour growth has been halted, but vision is poor because of retinal vascular occlusion in one case and optic atrophy in the other.

Our experience with radioactive plaque therapy started some 20 years ago with the then popular gamma-emitting radioactive cobalt plaque introduced by Stallard.²² Our experience was similar to that reported by McFall and Bedford,²³ namely that after an initially good result, vision was often lost from late radiation-induced vascular occlusion in the retina or optic nerve head. As a result of this early experience we gave up this form of therapy until the reportedly safer beta-emitting ruthenium/rhodium

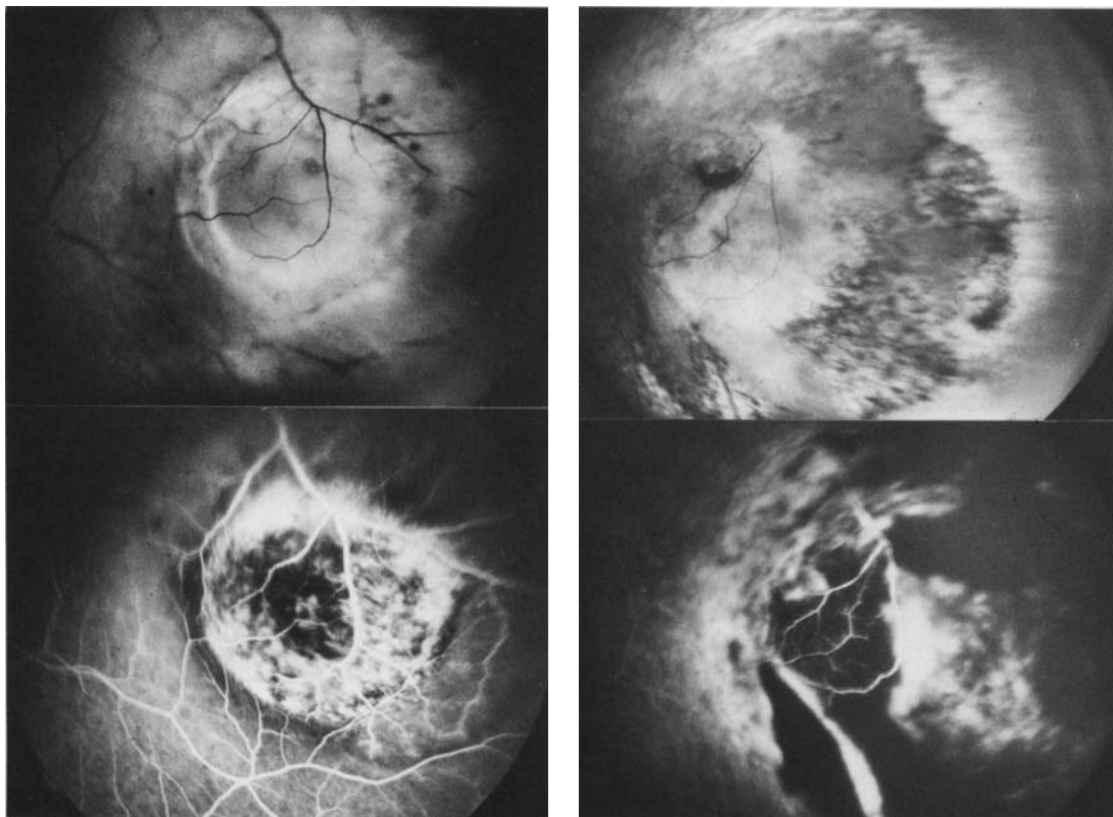


Figure 4: (a) Pre-treatment and (b) post-treatment appearance of a choroidal melanoma treated with ruthenium plaque. The radiation has caused considerable vascular closure in the choroid and in the overlying retina.

plaques became available to us. So far we have treated 15 patients with this form of radiation and although it is not without its problems the results in most cases have been good. Initially we aimed to give a dose of 80Gy (8000 rad) at the apex of the tumour, but later have modified this to give a larger dose at the base of the tumour, irrespective of tumour height, on the assumption (still to be tested) that beta-irradiation achieves its effect by virtue of vaso-occlusion of the choroidal vessels feeding the tumour (Figure 4). Complications of plaque therapy have included temporary choroidal and retinal detachment, retinal vascular occlusion, choroidal haemorrhage and diplopia. All these complications appear to be self-limiting, and most are compatible with retention of good vision. The effect on the tumour can be monitored by ultrasound and photography and, while a cessation of tumour growth is seen in all successfully treated cases, in only a proportion is there an actual regression of the tumour. Of 22 patients treated with radioactive plaque a good response was seen in all seven treated with cobalt, although further growth in the tumour occurred 15 years later in one patient. None of the cobalt-treated patients retained good vision. Of the 15 treated with ruthenium plaque, apart from one patient who died for reasons unconnected with the tumour, regression of tumour was seen in six and in seven growth of the tumour was halted. In only one case was there continued tumour growth.

When considering radiation we must not, of course, forget the possible use of charged particle therapy. Experience with protons or helium nuclei has been gained in Boston and in California^{24,25} and in suitable cases the results appear good, both in terms of visual outcome and subsequent mortality. Radiation damage to the retina or optic nerve can occur, however, and mortality rates are probably comparable to those for other methods of treatment. It is interesting that in one centre in the USA there has been a resurgence of interest in cobalt plaque therapy.

RESIDUAL/RECURRENT TUMOUR

With all the treatment modalities discussed there is the problem of recognition and management

of residual or recurrent tumour. Following local surgical resection, pigmented areas close to the resected area are commonly seen. Some of these are undoubtedly pigment epithelial proliferation and others, again, organized haemorrhage or collections of blood-containing macrophages. In some instances the residual lesion has the characteristics of the original tumour and undoubtedly represents residual tumour tissue. Occasionally, if inadequately dealt with, such tumours may grow to give a recurrence of the original lesion and one such patient has successfully undergone a second local resection and is alive and well seven years later. It has been our practice to treat any suspicious area as if it were residual tumour, using low-energy long-exposure laser for small flat areas of pigmentation and (occasionally) ruthenium plaque therapy for thicker lesions. Such therapy is usually effective in rendering any suspicious area flat and inactive. In 35 patients laser treatment of suspicious areas of pigmentation which may well not have been tumour was carried out, eight of the patients receiving treatment with ruthenium plaque in addition. In 30 of the 35 patients a satisfactory response to treatment was attained, while in 3 of the 35 treatment is as yet incomplete.

From what has been said, the treatment options obviously become fewer the larger the tumour. Small tumours of up to 10 mm in diameter may be managed by observation, laser, irradiation or surgical resection. Medium to large tumours of 10 to 15 mm in diameter may be best managed by local resection or possibly proton beam irradiation where this is available. Very large tumours still demand enucleation except in the case of only eyes where local surgical resection may be sight-saving in some cases. We tend to reserve surgical resection for tumours in the size range 10 to 15 mm in diameter. In some respects, smaller tumours are technically more difficult to resect surgically, and in any case they usually do well with less aggressive forms of management.

IMMUNE ASPECTS

One aspect we have not so far considered is the role of the immune system in relation to survival

after treatment of uveal melanoma. We and others have demonstrated both cell-mediated and humoral immunity to melanoma antigens in patients with these tumours, but as yet the relevance of this to survival is unknown.²⁶⁻²⁸ One of us (BD) has isolated a number of monoclonal antibodies to different surface components of melanoma cells and these are being tested for specificity against a bank of melanomas which we have established. Already, using a group of monoclonal antibodies, it is possible to identify those patients who have such antibodies in the circulation. A comparison of the immune status of these patients with subsequent survival statistics will be most useful. Future plans for these monoclonal antibodies include both a diagnostic and a therapeutic role, but their efficacy in either regard is still to be established.

CONCLUSION

The present conflict of views over the management of choroidal melanoma has led to the development of many different forms of treatment. The relative merits of these methods must eventually be assessed by prospective controlled trials, which may not be too difficult to arrange when both treatments under investigation involve retention of the eye. Where one of the options is enucleation, there are obvious difficulties in persuading patients to accept enucleation as a treatment based on random selection, and it may be that we will have to content ourselves with very accurate follow-up of carefully matched patients and tumours treated by enucleation or by some more conservative modality.

At present we believe it is useful for a few centres to gain experience in what is still unconventional management of potentially lethal tumours, and until we are more certain of survival statistics for each of the various options we would not recommend, for example, the wholesale adoption of local resection of choroidal tumours.

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“In the laboratory, the lowest levels of UV radiation (at specific wavelengths) that have been shown to be capable of producing direct UV photo damage to the ocular lens, while considerably lower than ambient (solar) UV radiation, are still significantly larger (at least by a factor of 10) than the UV radiation emission levels emitted by VDT’s (visual display terminals). Thus the best available current experimental and epidemiological evidence does not indicate that the level of UV radiation emitted by the VDT’s is capable of exerting any deleterious effects on the ocular lenses of personnel using these terminals”.

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